



**SIMATS Online Programming Lab**  
C PROGRAMMING

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QUESTIONS 25 / 25 completed

Average

**AVERAGE**

Write a C program to calculate the average of elements in an array.

**PROGRAM EDITOR** C Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,aver=0,div=1;
5     int a[]={5,5,2,2,1};
6     int size=sizeof(a)/sizeof(a[0]);
7     for (i=0;i<size;i++){
8         aver+=a[i];
9         div=aver/size;
10    }
11    printf("average of array=%d",div);
12    return 0;
13 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Reverse

**REVERSE**

Write a C program to reverse the elements of an array.

**PROGRAM EDITOR** C Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i;
5     int arr[5]={2,4,6,12,16};
6     for(i=1;i<5;i++)
7     {
8         printf("%d",arr[i]);
9     }
10    printf("\n");
11    {
12        for(i=4;i>0;i--)
13        {
14            printf("%d",arr[i]);
15        }
16    printf("\n");
17 }
```

INPUT

Your INPUT here!

QUESTIONS 25 / 25 completed

2nd Max

**2ND MAX**

Write a C program to find the second largest element in an array.

**PROGRAM EDITOR** C Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i;
5     int largest,second;
6     printf("enter the size: ");
7     scanf("%d",&n);
8     printf("enter elements:\n");
9     for(i=0;i<n;i++)
10         scanf
11
12 }
```

IN



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QUESTIONS 25 / 25 completed  
Sort A-Z  
**SORT A-Z**  
Write a C program to sort an array in ascending order.

PROGRAM EDITOR
C
Run Save

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10], n, i, j, temp;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    for(i = 0; i < n - 1; i++)
15    {
16        for(j = i + 1; j < n; j++)

```

INPUT  
Enter number of elements: 5  
Enter elements:  
40  
10  
30  
20  
50

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QUESTIONS 25 / 25 completed  
Equal  
**EQUAL**  
Write a C program to check if two arrays are equal or not.

PROGRAM EDITOR
C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int n,i,a[10],b[10],equal=1;
5     printf("enter the size:\n");
6     scanf("%d",&n);
7     printf("enter frist array:\n");
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10    printf("enter second array:\n");
11    for(i=0;i<n;i++)
12        scanf("%d",&b[i]);
13    for(i=0;i<n;i++)
14    {
15        if(a[i]!=b[i]){
16            equal=0;

```

INPUT  
1 2 3 4 5  
1 2 3 4 7



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QUESTIONS 25 / 25 completed  
Frequency  
**FREQUENCY**  
Write a C program to find the frequency of an element in an array.

PROGRAM EDITOR
C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int i,j;
5     int a[]={2,4,5,6,1,4,2,5,2,4};
6     int size=sizeof(a)/sizeof(a[0]);
7     int visit[]={0,0,0,0,0,0,0,0,0,0};
8     for(i=0;i<size;i++){
9         if(visit[i]==1)
10            continue;
11        int count=1;
12        for(j=i+1;j<size;j++)
13            if(a[i]==a[j])
14            {
15                visit[j]=1;
16                count++;

```

INPUT  
Your INPUT goes here!

Clipboard  
Item not Collected  
Format not supported by Office Clipboard

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QUESTIONS 25 / 25 completed

Duplicate

**DUPLICATE**

Write a C program to remove duplicate elements from an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j, k;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    for(i = 0; i < n; i++)
15    {
16        for(j = i + 1; j < n; j++)
```

INPUT

Enter number of elements: 7  
Enter elements:  
10  
20  
10  
30  
20  
40  
30

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QUESTIONS 25 / 25 completed

Insert

**INSERT**

Write a C program to insert an element at a specific position in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[100],n,i,pos,value;
5     printf("enter the number of elements:");
6     scanf("%d",&n);
7     printf("enter the array:");
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10    printf("enter the position:");
11    scanf("%d",&pos);
12    printf("enter value:");
13    scanf("%d",&value);
14    for(i=n;i>=pos;i--)
15        a[i]=a[i-1];
16    a[pos-1]=value;
```

INPUT

5  
10 20 30 40 50  
3  
25

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QUESTIONS 25 / 25 completed

Min

**MIN**

Write a C program to find the smallest element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,min;
5     int a[5]={2,4,6,6,9};
6     min=a[0];
7     for (i=1;i<5;i++)
8     {
9         if(a[i]<min)
10             min=a[i];
11     }
12     printf("min=%d",min);
13     return 0;
14 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Max-Min

**MAX-MIN**  
Write a C program to find the maximum and minimum element in an array.

**PROGRAM EDITOR**

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,max,min;
5     int a[5]={2,4,5,7,8};
6     max=min=a[0];
7     for (i=0;i<5;i++)
8     {
9         if (a[i]>max)
10            max=a[i];
11        if(a[i]<min)
12            min=a[i];
13    }
14    printf("max=%d",max);
15    printf("min=%d",min);
16    return 0;
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Count E-O

**COUNT E-O**  
Write a C program to count the number of even and odd elements in an array.

**PROGRAM EDITOR**

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,even=0,odd=0;
5     int a[6]={2,4,7,9,6};
6     for(i=0;i<5;i++)
7     {
8         if(a[i]%2==0)
9             even++;
10        else
11            odd++;
12    }
13    printf("even count=%d\n",even);
14    printf("odd count =%d\n",odd);
15    return 0;
16 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Sum E-O

**SUM E-O**  
Write a C program to find the sum of elements at even and odd positions in an array.

**PROGRAM EDITOR**

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,even=0,odd=0;
5     int a[5]={2,8,9,5,7};
6     for(i=0;i<5;i++)
7     {
8         if(a[i]%2==0)
9             even+=a[i];
10        else
11            odd+=a[i];
12    }
13    printf("even sum =%d\n",even);
14    printf("odd sum=%d\n",odd);
15    return 0;
16 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Product

**PRODUCT**  
  
Write a C program to find the product of elements in an array.

**PROGRAM EDITOR** C Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[100],n,i,product=1;
5     printf("enter the number of elements:\n");
6     scanf("%d",&n);
7     printf("enter array of elements:\n");
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10    for(i=0;i<n;i++)
11        product=product*a[i];
12    printf("product =%d",product);
13    return 0;
14 }
```

INPUT

```
5
2 3 4 5 6
```



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QUESTIONS 25 / 25 completed

Fun-Min-Max

**FUN-MIN-MAX**  
  
Write a C program to find the largest and smallest elements in an array using functions.

**PROGRAM EDITOR** C Run Save

```
1 #include<stdio.h>
2 int largest (int a[],int n)
3 {
4     int i,max=a[0];
5     for(i=1;i<n;i++)
6     {
7         if(a[i]>max)
8             max=a[i];
9     }
10 int smallest (int a[],int n)
11 {
12     int i , min=a[0];
13     for(i=0;i<n;i++)
14     {
15         if(a[i]<min)
16             min=a[i];
17     }
18 }
```

INPUT

```
5
12 8 25 4 18
```



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QUESTIONS 25 / 25 completed

2nd Min

**2ND MIN**  
  
Write a C program to find the second smallest element in an array.

**PROGRAM EDITOR** C Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,max;
5     printf("enter the number:");
6     scanf("%d",&n);
7     printf("enter elements");
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10    max=a[0];
11    for(i=1;i<n;i++)
12        if(a[i]>max)
13            max=a[i];
14    printf("largest=%d",max);
15    return ;
16 }
```

INPUT

```
5
10 25 7 40 15
```





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**QUESTIONS** 25 / 25 completed

Delete

**DELETE**  
Write a C program to delete an element from an array at a specific position.

**PROGRAM EDITOR**
C

Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[100],n,i,pos;
5     printf("enter the number of elements:\n");
6     scanf("%d",&n);
7     printf("enter array elements:\n");
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10    printf("enter the position to delete:\n");
11    scanf("%d",&pos);
12    for(i=pos-1;i<n-1;i++)
13        a[i]=a[i+1];
14    n--;
15    printf("array after the deletion:\n");
16    for(i=0;i<n;i++)

```

**INPUT**  
5  
10 20 30 40 50  
3



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**QUESTIONS** 25 / 25 completed

Merge

**MERGE**  
Write a C program to merge two sorted arrays into a single sorted array.

**PROGRAM EDITOR**
C

Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int i;
5     int a[20]={2,10,9,15,6};
6     int b[5]={3,6,7,1,9};
7     int sizea= 5,sizeb=5;
8     for(i=0;i<sizeb;i++)
9         a[sizea++]=b[i];
10    for(i=0;i<sizea;i++)
11        printf("%d ",a[i]);
12    return 0;
13 }

```

**INPUT**  
Your INPUT goes here!



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**QUESTIONS** 25 / 25 completed

Max

**MAX**  
Write a C program to find the largest element in an array.

**PROGRAM EDITOR**
C

Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[5],i,max;
5     printf("enter 5 elements:\n");
6     for(i=0;i<5;i++)
7     {
8         scanf("%d",&a[i]);
9     }
10    max=a[0];
11    for(i=1;i<5;i++)
12    {
13        if(a[i]>max)
14        {
15            max=a[i];
16        }

```

**INPUT**  
10 15 20 25 30



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QUESTIONS 25 / 25 completed

Unique

**UNIQUE**  
  
Write a C program to print all the unique elements in an array.

**PROGRAM EDITOR**

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j, count;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Unique elements:\n");
15
16    for(i = 0; i < n; i++)
```

**INPUT**  
  
Enter number of elements: 7  
Enter elements:  
10  
20  
10  
30  
40  
20  
50



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QUESTIONS 25 / 25 completed

Rotate

**ROTATE**  
  
Write a C program to rotate an array to the left by n positions.

**PROGRAM EDITOR**

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, r, i, j, temp;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter number of positions: ");
15    scanf("%d", &r);
16
```

**INPUT**  
  
Enter number of elements: 5  
Enter elements:  
10  
20  
30  
40  
50  
Enter number of positions: 2



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QUESTIONS 25 / 25 completed

Check Sort

**CHECK SORT**

Write a C program to check if an array is sorted in ascending or descending order.

PROGRAM EDITOR

C

RunSave

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i;
6     int asc = 1, desc = 1;
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11    printf("Enter elements:\n");
12    for(i = 0; i < n; i++)
13        scanf("%d", &a[i]);
14
15    for(i = 0; i < n - 1; i++)
16    {
```

INPUT

Enter number of elements: 5

Enter elements:

10

20

30

40

50



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QUESTIONS 25 / 25 completed

Median

**MEDIAN**

Write a C program to find the median of an array.

PROGRAM EDITOR

C

RunSave

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j, temp;
6     float median;
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11    printf("Enter elements:\n");
12    for(i = 0; i < n; i++)
13        scanf("%d", &a[i]);
14
15    for(i = 0; i < n - 1; i++)
16    {
```

INPUT

Enter number of elements: 5

Enter elements:

10

30

20

50

40